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Infrastructure Services and Early Childhood Development in Latin America and the Caribbean: Water, Sanitation, and Garbage Collection[☆]

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ABSTRACT

Access to essential infrastructure services such as water, sanitation, and garbage collection can considerably affect children's environment and may play a significant role in shaping early childhood developmental and health outcomes. Using data from the Multiple Indicator Cluster Surveys (MICS) and the Demographic and Health Surveys (DHS) for 18 countries in Latin America and the Caribbean (LAC), we show a significant positive association between access to water and sanitation and early childhood development, as well as reduced instances of stunting. In addition, we identify a negative association between access to improved garbage collection services and the rates of stunting among children under five. Our findings are robust after using alternative measures for access and controlling for individual, maternal, and household factors, alongside considerations of household wealth, caregiver's stimulation activities and behaviors, and local community spillovers. Similarly, the economic relevance of the relationship is highlighted by the substantial gap relative to the size of the vulnerable groups, persisting even after adjusting for confounding variables. Our results also suggest that households may be able to lessen the potential impact of pollutants through mitigation measures such as treating water to make it safe for consumption, using handwashing cleansers, and storing household trash in lidded containers. The current findings underscore the importance of investing in basic infrastructure services as a critical component of comprehensive strategies to enhance early childhood development and health in low- and middle-income countries. We emphasize the importance of considering the quality and type of infrastructure services alongside their availability. Future research should incorporate more complete and detailed data to improve understanding of the causal relationship between water, sanitation, and garbage collection and early childhood development, as well as the mechanisms underlying the observed associations.

1. Introduction

More than 250 million children younger than five living in low- and middle-income countries are at risk of not reaching their developmental potential due to adversity and lack of nurturing care (i.e., good health, adequate nutrition, early learning opportunities, safety and security, and responsive caregiving) (Black et al., 2017; Lu et al., 2020; McCoy, Seiden, Cuartas, Pisani, & Waldman, 2022). This is concerning as early childhood development (ECD) predicts educational trajectories, future health, and economic and social well-being throughout an individual's

lifespan (Black et al., 2017; Heckman, 2006). Indeed, brain development and the acquisition of foundational skills are highly sensitive to environmental input early in life (Sheridan & Nelson, 2009). Early experiences — positive or negative — thus have significant long-term consequences for individual and societal welfare (Black et al., 2017; Cunha & Heckman, 2007; Knudsen, Heckman, Cameron, & Shonkoff, 2006).

Extensive evidence from multiple disciplines has shown that young children are particularly sensitive to various nutritional, health, and psychosocial factors, including the home environment, access to learning opportunities, and exposure to violence (Attanasio, Cattani,

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Fitzsimons, Meghir, & Rubio-Codina, 2020; Jeong, McCoy, & Fink, 2017; McCoy, Roy, & Raver, 2016; Salhi et al., 2021; Wolf & McCoy, 2019). While there is extensive literature within the field of public health and pediatrics assessing the impact of the physical environment (in particular, the relation between water, sanitation, and hygiene with nutrition) on ECD (Danaei et al., 2016; Sania et al., 2019), the bulk of this research has focused on stunting as an outcome and not child development specifically. Stunting has often been used as a proxy for child development, though recent evidence suggests that it is an inadequate proxy (Perumal, Bassani, & Roth, 2018; Sania et al., 2019). In this context, relatively little effort has been made to integrate developmental science with environmental health (Trentacosta, Davis-Kean, Mitchell, Hyde, & Dolinoy, 2016). This work is nonetheless important as climate change, industrialization, and lack of access to basic infrastructure have increased several specific environmental risks that may be particularly influential early in life. These are especially pronounced in low- and middle-income countries (LMICs), where millions of children lack access to clean water sources, improved sanitation, and adequate waste management (McCoy et al., 2022). Indeed, poor water, sanitation, and garbage disposal represent particular risks for children as they consume more water, food, and air relative to their body weight than adults, while their metabolism is less capable of coping with contaminants and their developing brain is more sensitive to pollutants (Trentacosta et al., 2016).

Inadequate or polluted water, poor sanitation, and garbage disposal can affect ECD through at least three mechanisms. First, poor hygiene, polluted water, and inadequate sanitation and waste management can lead to biological changes affecting the metabolic system with downstream consequences on children's weight and nutrition (Heindel & Schug, 2014). Second, ingesting pollutants and microbial contaminants can disrupt the normal development and pruning of synapses in key areas of the brain (e.g., the prefrontal cortex), shaping brain structure and function and consequently cognition, social-emotional development, and behavior (Walker et al., 2011). Interventions combining water, sanitation, handwashing, and nutrition might improve child motor development (Stewart et al., 2018).¹ Finally, pollution and the lack of clean water, adequate sanitation, and waste management sources can be a major source of stress for families, which can affect their behavior and the type and quality of care that they provide to children (Bornstein, Leventahl, & Lerner, 2015). In sum, evidence supports hypothesizing that inadequate water, sanitation, and garbage disposal are linked to childhood health and development.

Consistent with these findings, investing in improved water, sanitation, and waste management infrastructure can be a key strategy for promoting positive health and developmental outcomes early in life (Acuff & Kaffine, 2013). Environmental problems such as odors and air and water pollution are caused by poor waste management and may be exacerbated as the climate crisis deepens (Janmaimool, 2017; Vaccari, Tudor, & Vinti, 2019). Poor waste management increases the risk of severe diseases, like dysentery and jaundice (Rathnamala, Ashwini, & Babitha, 2021). However, budget and financial constraints, socioeconomic factors, and negative attitudes towards waste management make it difficult to invest in waste management infrastructure in LMICs (Al-Khatib, Abu Fkhidah, Khatib, & Kontogianni, 2016; Keser, Duzgun, & Aksoy, 2012; Manga, Oru, & Ngwabie, 2019; Rathnamala et al., 2021; Schiopu, Apostol, Hodoreanu, & Gavrilescu, 2007; Vaccari et al., 2019; Wang, Cheng, Reisner, & Liu, 2018). Furthermore, these policies are hard to target as responsibility for the generation of solid waste is shared across industry, households, and government (Abdulredha et al., 2018; Agyei-Mensah & Oteng-Ababio, 2012; Hamilton, Sproul, Sunding, & Zilberman, 2013; Yamamoto & Kinman, 2022).

¹ However, modestly improved sanitation practices do not necessarily enhance child health outcomes (Patil et al., 2014). This shows the complexity of achieving health benefits through interventions.

Addressing environmental infrastructure gaps is crucial for enhancing ECD in the Latin America and the Caribbean (LAC) region (Günther & Lahoti, 2021; Korc & Hauchman, 2021; Schady, 2015). While LAC has made significant strides in enhancing access to clean water, sanitation remains a pressing concern, especially in rural areas (Alvarez, 2019; Jeuland, Fuente, Ozdemir, Allaire, & Whittington, 2013). Similarly, the region has made significant progress in waste management, with an average solid waste collection rate higher than the global average. Yet, challenges persist in achieving consistent and equitable waste management services across different areas. The region grapples with variations in service quality and frequency, as well as a low level of municipal planning in solid waste management (Grau, Terraza, Velosa, Rihm, & Sturzenegger, 2015). These infrastructural challenges not only pose immediate health risks but also could have long-term implications on cognitive and physical development. The region's unique socio-economic dynamics, coupled with its diverse topography and climate, further accentuate these challenges. Vulnerable areas often are characterized by limited state presence and investment. Families in these locations often miss the resources or knowledge to adopt mitigative measures, exacerbating the health and developmental risks for children. In stark contrast, urban centers, despite better infrastructure, grapple with challenges posed by rapid urbanization and population density, often leading to overburdened facilities (Sparkman & Sturzenegger, 2016).

This paper aims to provide evidence on the associations between improved water, sanitation, and waste management infrastructure with ECD in LAC. Our findings can provide important evidence to support targeted investments that benefit the environment, local populations, and young children. In particular, we seek to address the following research questions. First, does access to improved water, sanitation, and waste management correlate with young children's health and development? Second, to what extent do the links between access to improved water, sanitation, and waste disposal vary across countries, when controlling for individual and household characteristics, such as maternal education, wealth, health status and health practices, stimulation behaviors, local community spillovers, and the availability of certain assets and goods (e.g., hand cleansers, refrigerators, access to bottled water, and lidded containers)?

We use two main data sources to examine these questions. First, we employ UNICEF's Multiple Indicators Cluster Surveys (MICS) from Belize, Costa Rica, Cuba, El Salvador, Guyana, Honduras, Jamaica, Panama, Paraguay, Dominican Republic, Saint Lucia, Suriname, Trinidad and Tobago, and Uruguay, which include information on access to improved water and sanitation and on child health and development. We also use the Demographic and Health Surveys (DHS) from Bolivia, Colombia, Dominican Republic, Guyana, Nicaragua, and Peru, which provide information on child health and nutrition, and household waste management. We conducted a thorough multivariate statistical analysis, utilizing linear probability models to explore the correlations between infrastructure access and developmental and health outcomes, while taking advantage of the differences in infrastructure between countries. To mitigate potential endogeneity concerns, especially the confounding effect of household wealth, individual behaviors, and local community spillovers, we incorporated control variables in our models. We further validated our findings by incorporating the stimulation index, which is a widely recognized predictor of ECD. Additionally, we used categorical variables to analyze access to water and sanitation, and applied a propensity score matching approach to the data. Despite these variations, our results remain consistent. While we recognize these potential endogeneity issues and cannot fully address them, we do not claim causal effects but rather demonstrate a robust and consistent link between infrastructure access and early childhood development and health outcomes.

The findings of our paper significantly emphasize the core associations between access to improved water, sanitation, waste management, and the developmental and health outcomes of young children.

Children in households with better access to these facilities are more likely to be developmentally on track and less prone to stunting. Notable variability in these associations across countries calls for further exploration of other environmental factors affecting children's health and development. Additionally, improved garbage collection services are linked with reduced rates of stunting among children under five. We also explore mitigating behaviors that households can adopt amid inadequate access to clean water and sanitation facilities. Our findings advocate for promoting household water treatment practices and using lidded containers for waste as provisional solutions toward universal access to water and sanitation. Examining predicted rates of stunting among children who have experienced recent illness episodes highlights potential underlying mechanisms affecting child development, emphasizing a potential health-related mechanism.

Our findings resonate with and extend the existing literature, marking an advancement in comprehending the infrastructure-health nexus. Randomized trials have shown that water, sanitation, and hygiene interventions often impact diarrhea incidence and infections but do not have the expected effects on child stunting and other health outcomes (Cumming et al., 2019; Freeman et al., 2017). Yet, large-scale observational studies often find more consistent positive effects of access to adequate water, sanitation, and garbage collection on health outcomes. For example, Gao et al. (2021) used a sample of more than 88,000 children aged 26–59 months in 20 countries in Sub-Saharan Africa and found a positive association between a composite index of improved housing (including access to better water sources and sanitation facilities) and early childhood development. An study by Headey and Palloni (2019) used representative data from 59 countries and a difference-in-difference approach and found that access to improved water was not associated with health outcomes, except for access to water piped into the home, which predicted reductions in stunting. Moreover, improved sanitation predicted reductions in diarrhea prevalence and mortality but not stunting. In addition, Schady (2015) highlighted the disparities in water and sanitation infrastructure access in LAC. The public health implications of inadequate facilities, including the link between microbial ingestion and health adversities such as diarrhea, help to explain the high mortality rates in children under five years old in developing regions.

While existing evidence is somewhat mixed, but substantiates the importance of improved infrastructure for child health outcomes, the literature has focused exclusively on health outcomes and overlooked developmental outcomes, which are critical for life long trajectories. Therefore, there is a need to conduct new evaluations to assess the impact on child morbidity, nutritional status, and development. There is a lack of well-identified evidence within the LAC region, advocating for comprehensive evaluations to understand the multi-dimensional benefits of improved water and sanitation access for children over time (Günther & Lahoti, 2021; Schady, 2015). To the best of our knowledge, our paper is the first to explore the entangled relationships between this set of types of infrastructure access — most notably waste management, but also water and sanitation — and their link to developmental and health outcomes in LAC, leveraging between-country heterogeneity. We emphasize the importance of considering the quality and type of infrastructure services alongside their availability. The literature has often focused on water and sanitation, but the inclusion of complementary services, such as waste management, as part of basic infrastructure is crucial for a holistic approach. This disaggregated information is critical for understanding specific early childhood outcomes that may be impacted by physical environment features and for informing future investments in infrastructure to promote positive developmental trajectories from a young age.

The remainder of this paper is organized as follows. In Section 2 we summarize the methodology, including the measurement tools and estimation method. Section 3 describes the data used to assess the links between access to improved water sources, sanitation, and garbage disposal and early childhood developmental and health outcomes. Section 4 summarizes the results. We conclude with a general discussion and implications for future research and practice.

2. Methodology: Measurement and estimation

In this paper, we consider two main outcome variables: (1) child development and (2) child health/nutrition. We use the early childhood development index (ECDI) (Loizillon, Petrowski, Britto, & Cappa, 2017) as a measure of young children's development. The ECDI is a 10-item parent-reported scale that aims to assess the basic developmental skills of children aged 3–5 years in four developmental domains: social-emotional, literacy-numeracy, approaches to learning, and physical. The social-emotional domain consists of three items asking whether the child (1) gets along well with other children, (2) kicks, bites, or hits other children or adults, and (3) is easily distracted. Two items cover literacy-numeracy: (1) whether the child can identify or name at least ten letters of the alphabet, and (2) whether the child can read at least four simple words. Approaches to learning is measured by two items: (1) whether the child follows simple instructions and (2) whether, when given something to do, the child is able to do it independently. Finally, the physical domain includes the following two items: (1) whether the child can pick up a small object with two fingers and (2) whether the child is often too ill to play.

The ECDI is one of the few comparable measures of child development used globally and, specifically, across LAC. It serves as an important policy indicator, as the lack of comparable child development indicators has constituted a barrier to monitoring progress, ensuring accountability, and investing effectively in ECD (Engle et al., 2007).² The ECDI is highly policy-relevant as it is an interim measure for reporting on Sustainable Development Goal indicator 4.2.1 (UNICEF et al., 2020). This indicator measures the percentage of children who are developmentally on track in health, learning, and psychosocial well-being (United Nations, 2015). Following UNICEF guidelines (Loizillon et al., 2017) and prior research (Lu et al., 2020), we consider a child to be “on track” if they meet the criteria for being developmentally on track in at least three of the four domains of the ECDI. The domains include literacy-numeracy, physical, social-emotional, and learning. Fig. 1 shows the percentage of children with ECDI on track for all MICS rounds in LAC.

To measure child health, we use information on children's anthropometry, using Shorr measuring boards for height. We follow World Health Organization standards (WHO, 2009) and classify children aged 0–5 years with a height-for-age z-score (HAZ) lower than 2 standard deviations below the median score as stunted.³ Fig. 2 shows how these indicators have progressed in LAC over the past two decades.

Our independent variables of interest are access to adequate water, sanitation, and garbage disposal in the dwelling. Children are considered to have access to adequate water if their main source of drinking water comes from a protected and treated source. This includes water piped directly into the dwelling, yard, or plot, public taps or standpipes, tubewells or boreholes, protected wells, protected springs, and treated water from public water treatment plants. Additionally, packaged bottled water and water from water kiosks also are considered adequate sources.⁴ Tanker-trucks, carts with small tanks or drums, and other packaged water like sachet water are also considered reliable sources

² Many other global population measures of ECD exist, such as the Assessment of Early Childhood Development (AIM-ECD), the Model of Early Learning and Development (MODEL), the Caregiver-Reported Early Development Instruments (CREDI), and the Global Scale for Early Development (GSED). In this paper, we focus on ECDI only.

³ As mentioned previously, nutritional factors such as stunting are conceptualized as inputs to child development, not outcomes of child development itself (Danaei et al., 2016; Sania et al., 2019). However, they are considered outcomes in this analysis to comprehensively assess the links with child health and development.

⁴ In the case of Panama, specific water treatment plant options have been recoded to be included under piped water, acknowledging them as treated water options.

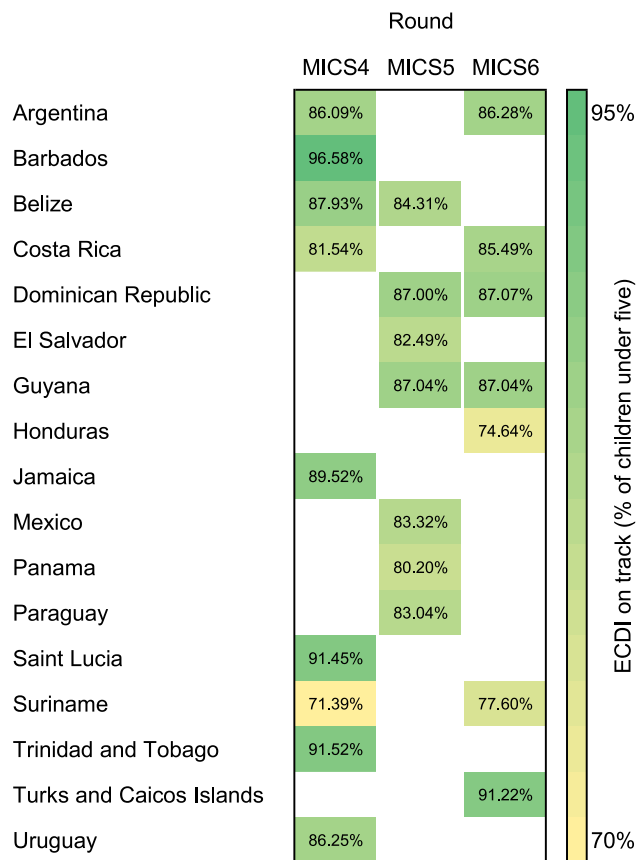


Fig. 1. Percentage of children with early childhood development index on track in LAC.

Notes: This heatmap visualizes the percentage of children with early childhood development index (ECDI) on track across various countries in Latin America and the Caribbean (LAC), as captured in different rounds of the Multiple Indicator Cluster Surveys (MICS). The rounds 4, 5, and 6 correspond to three different periods: 2009–2012, 2012–2016, and 2017–2019, respectively. The colors represent the percentage of children on track, with darker shades indicating higher percentages. Each column corresponds to a different MICS round, while each row represents a specific country. The percentages presented are weighted statistics calculated at the child level to ensure national representativeness. These statistics may vary from the ones in Sections 3 and 4 as they do not incorporate such weighting and final samples differ for data analysis purposes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of safe drinking water (WHO, 2019). On the other hand, sources such as unprotected wells and springs, rainwater collection, surface water from rivers, dams, and lakes, and other unspecified or unknown sources are categorized as inadequate. Regarding sanitation, children are considered to have access to adequate sanitation if their dwellings have private sanitation facilities such as flush or pour-flush toilets connected to a piped sewer system or septic tank, various types of pit latrines (including ventilated improved pit latrines and those with slabs), and composting toilets. Inadequate sanitation includes the use of shared or public-use facilities, flush or pour-flush toilets that dispose waste elsewhere, pit latrines without slabs or open pits, hanging latrines, and practices of open defecation. Last, we also add garbage collection, including details about household storage practices and quality of service. In general, non-collected garbage disposal includes open dumping or burning of waste.

Finally, the data include demographic, socioeconomic, and contextual information that we use as covariates in our models. This includes information on children's age, sex, and health; mothers' education; household composition; individual practices and behaviors; and wealth quintiles, as measured by a principal component index of basic dwelling

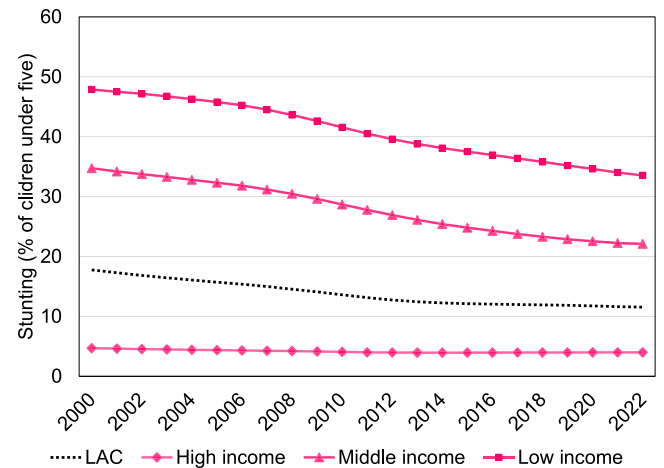


Fig. 2. Trends in stunting: LAC and global income groups.

Notes: This figure illustrates the progression of stunting prevalence among children under five years old in the Latin America and Caribbean (LAC) region against global income groups, as defined by the World Bank. It delineates the percentage of children exhibiting stunting, characterized by a height for age more than two standard deviations below the median of the international reference population. Simultaneously, the figure highlights existing nutritional disparities and trends over time. The data for this visualization are sourced from the World Development Indicators.

characteristics and asset ownership.⁵ In addition, we use a stimulation index as a control variable, which summarizes caregivers' engagement in six different play/learning activities with children, following prior research showing associations between these activities and young children's cognitive, social-emotional, and motor development (Cuartas et al., 2023).⁶ The index is a categorical variable ranging from 0 to 6, based on whether anyone in the household engaged in these activities with the child in the last three days. It includes stimulation from any caregiver, not just the mother, thus capturing a broader range of interactions. Specifically, the MICS asks whether anyone in the household engaged in the activities like reading, telling stories, singing songs, taking the child outside, playing, and spending time naming, counting, or drawing things with the child. The responses are recoded so that a higher score indicates greater levels of stimulation.

In our empirical analysis, we employ a linear probability model to investigate the relationship between access to basic infrastructure services and child health and developmental outcomes. The model is specified as follows:

$$y_{ict} = \beta_0 + \beta_1 \cdot \text{access}_{ict} + \gamma_{ctr} + \mathbf{X}_{ict}\beta + \epsilon_{ict}$$

where y_{ict} represents the developmental and health outcome of interest for child i in country c and year t . These outcomes encompass binary indicators for ECDI and stunting. access_{ict} is a binary variable capturing access to essential infrastructure, including water, sanitation, and garbage collection, for child i in country c and year t . γ_{ctr} denotes country-year-region fixed effects, controlling for unobserved heterogeneity across regions within countries over time. $\mathbf{X}_{ict}$ is a vector of covariates incorporating individual characteristics of the child, maternal characteristics, and household characteristics. β represents the vector of coefficients associated with the covariates and ϵ_{ict} is the error term. We cluster the standard errors at the community level using each survey's primary sampling units (PSU) to account for within-group

⁵ We do not directly calculate the wealth index as it is already accessible in both the MICS and DHS data sets. See Appendix Table A2 for additional details.

⁶ The inclusion of a stimulation covariate is informed by evidence that caregivers' stimulation significantly impacts early child development. See Cuartas et al. (2023) and Jeong, McCoy, Yousafzai, Salhi, and Fink (2016).

correlation, recognizing that observations within the same PSU may not be independent and thus could exhibit correlation in their unobserved influences on the health and developmental outcomes.

This model allows us to estimate the average correlation of having access to infrastructure on children's developmental and health outcomes, accounting for observed characteristics and fixed effects. We include additional controls in various model specifications to mitigate potential endogeneity issues and to provide robustness checks. A comprehensive set of additional variables is included in our analysis to account for various factors that could influence child development and health outcomes (i.e., confounders). These controls include child characteristics (age, sex, health status), maternal characteristics (education, health behaviors), household factors (wealth, household composition), and environmental factors (stimulation index). The inclusion of these controls helps isolate the link of infrastructure access by accounting for other variables that could confound the results.

While we recognize the potential endogeneity issues in our paper, we strive to address these concerns through several approaches. Endogeneity concerns stem from both infrastructural supply-side factors, such as government-provided services, and behavioral demand-side factors, such as household preferences and knowledge. For example, households with better access to water, sanitation, and garbage collection services might also benefit from other unobserved amenities like superior health services, better schools, or more stable electricity supply. Similarly, households that demand better infrastructure services might have higher hygiene standards, better nutrition, and overall healthier environments, which are difficult to observe and control for fully.

To mitigate these issues, we could incorporate community-level variables related to infrastructure, such as the percentage of children born in medical facilities and the share of households with electricity. Additional spillovers come from the access to health services, including medical facility births, antenatal care, and vaccinations. We use country-year-region fixed effects to control for unobserved heterogeneity across regions within countries over time. Primarily, these fixed effects help control for community-level variations, but to enhance accuracy, we also test the results using primary sampling unit (PSU) fixed effects. Additionally, to account for household-level behaviors and characteristics that could influence the outcomes, we include controls for recent illness episodes, vaccination behavior, health and nutritional status, and hygiene practices.

Furthermore, we employ a propensity score matching approach to strengthen our analysis, although we acknowledge that matching cannot account for all unobservable factors. Recognizing the limitations, we do not claim causal effects but rather demonstrate a robust and consistent link between infrastructure access and early childhood development and health outcomes. Our findings highlight associations and underscore the importance of further research to establish causality.

3. Data

To assess the links between crucial basic services such as water, sanitation, and garbage collection and ECD and child health, we employ data from the Multiple Indicators Cluster Survey (MICS) and the Demographic and Health Surveys (DHS). In total, we use data for 18 countries and 36 surveys for the period 1991–2019 — see Figure A1 for the geographical coverage of our analyses and Appendix Tables A1 and A2 for sample sizes in each country and outcome, and variable definitions, respectively.

The MICS is a household survey designed to provide comparable information on the well-being of children and women, primarily in low- and middle-income countries. All MICS surveys follow a two-stage cluster sampling protocol, based on probabilistic, random samples of households with children under the age of five from enumeration areas drawn from national censuses. These surveys include questionnaires about the household (e.g., dwelling characteristics and access to goods

and services), women (e.g., background characteristics and maternal health), and children under five (e.g., development, and learning environment).

We use MICS data from a total of 22 surveys in 14 countries, although we use different samples to study different outcomes. This approach is adopted due to the substantial variation in the number of observations when comparing developmental outcomes with health outcomes across the datasets. Appendix Table A3 presents descriptive statistics that offer a general overview of the ECDI as well as data on access to improved water sources and sanitation infrastructure for a sample of 19 surveys with almost 38,000 child–mother dyads. We find that, on average, 83% of children are on track in overall development, with 28% on track in literacy/numeracy, 99% in physical development, 82% in socio-emotional development, and 97% in learning.⁷ Access to sanitation and water infrastructure services is 91% and 90%, respectively. The average child in the ECDI sample is 48 months old and belongs to a household of five to six members, with one to two children under the age of five. The gender distribution in the sample is relatively balanced, with female children comprising 49% of the sample.

The sample used to analyze health outcomes consists of around 67,000 child–mother dyads in 17 surveys.⁸ The statistics indicate a 12% incidence of stunting among children under the age of five with 87% access to water and 90% access to sanitation services. These figures suggest that although many children in the study region are developing as expected according to the ECDI, significant challenges remain regarding critical developmental skills, health outcomes, and access to infrastructure services.

There are significant disparities between countries in access to these infrastructure services for households with children under five.⁹ While MICS data indicate high overall level of access to these services in LAC, a substantial number of households with children under five still do not have access. Notably, Guyana, Panama, Paraguay, and Suriname lag far behind the region in terms of access to water and sanitation, while Uruguay and Jamaica perform significantly better. This variation in access between countries is crucial for our paper as it provides sufficient variance of infrastructure access for our analyses. Thus, the use of multiple surveys allows us to capture the full range of variation in access to water and sanitation services, despite the high overall rates of access in the region.

In addition, the country data also reveal some patterns in access to water and sanitation services in each country. For instance, in Cuba, while access to water is almost universal, about 10% of households with children under five still lack access to adequate sanitation services. Access to sanitation services is likewise significantly lower than access to water services in Panama. These nuances in access to different types of infrastructure services are crucial to consider when designing policies and interventions to improve the well-being of children and women in the region. Although access to water and sanitation services is highly correlated, this does not necessarily mean these services are always connected. In some cases, households may have access to clean water but not adequate sanitation facilities, and vice versa. On average, 3% of children lack access to both water and sanitation, 13% lack access to one of them, and 84% have access to both.

Fig. 3 presents a detailed analysis of the relationship between access to water and sanitation services and early childhood development and health, as measured by two distinct indicators (one per panel): (a) the overall ECDI on track and (b) stunting. In each panel, the indicator is compared between children who have access to both water and sanitation services and those who do not.¹⁰

⁷ Please note that all statistics in Sections Section 3 and beyond are unweighted. Differences with national representative statistics may exist.

⁸ Additional information on the sample is provided in Appendix Table A4.

⁹ Appendix Table A5 shows access to water and sanitation services by country using the ECDI sample. Statistics are not significantly different when using the health sample.

¹⁰ As mentioned before, the number of surveys included for each indicator varies due to differences in data collection across surveys.

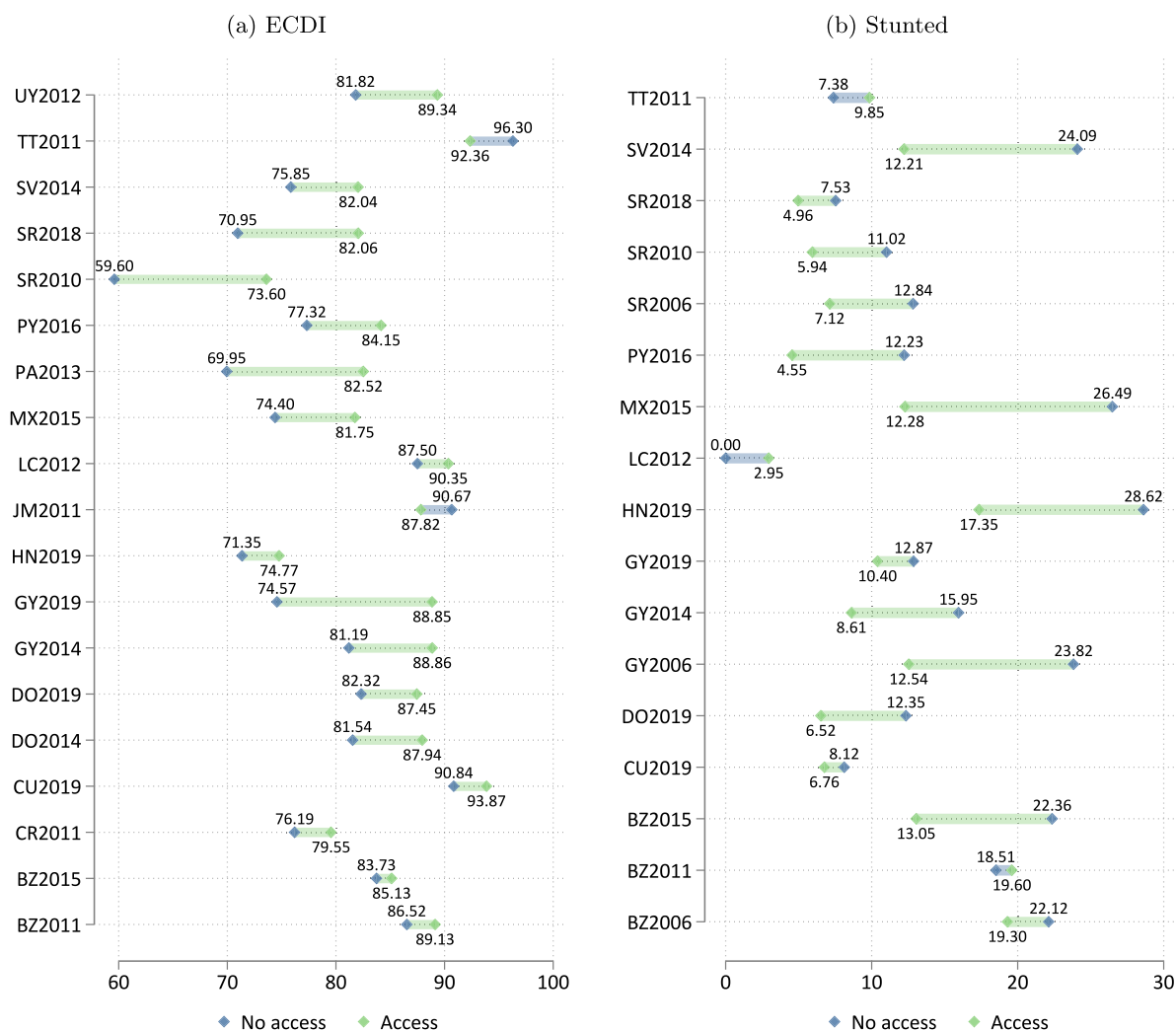


Fig. 3. Developmental and health outcomes by survey and access to both water and sanitation.

Notes: This chart displays the percentage of children who are considered developmentally on track according to the early childhood development index (ECDI), as well as the rates of stunting. Figures for each of these categories are shown in Figs. 3(a), and 3(b), respectively. Statistics are unweighted averages calculated separately for each survey for children with and without access to infrastructure services. Data were obtained from the Multiple Indicators Cluster Survey (MICS) for Belize (BZ), Costa Rica (CR), Cuba (CU), Dominican Republic (DO), El Salvador (SV), Guyana (GY), Honduras (HN), Jamaica (JM), Mexico (MX), Panama (PA), Paraguay (PY), Saint Lucia (LC), Suriname (SR), Trinidad and Tobago (TT), and Uruguay (UY).

Panel A shows that in most countries, children with access to water and sanitation services are more likely to be on track developmentally as measured by the ECDI compared to those without access. For instance, in countries like Guyana, Mexico, Panama, and Suriname, a large difference in early childhood development between children with and without infrastructure access is visible. Suriname has the lowest percent of children with ECDI on track compared to all other groups, which is concerning. In 2010, this percentage was at 60% for children without access. However, this figure has improved significantly since then, as of 2018. Similar patterns are observed for sanitation access, with a higher percentage of children with access to sanitation being on track with ECDI in most countries. In nearly all countries, the difference in the percentage of children who are developmentally on track is significant between those who do and do not have access.

The data also show in Panel B that the percentage of children with stunting is higher among those without access to infrastructure in most countries, except Trinidad and Tobago and Saint Lucia. The largest absolute difference is found in Honduras, Mexico and El Salvador, between those with and without access, while Guyana in 2019 has the smallest difference. Overall, the results presented in Fig. 3 indicate that access to water and sanitation services might be linked to early childhood development outcomes, with children who have access to

these services generally having better ECDI scores and experiencing lower rates of stunting.

We employ data from the Demographic and Health Surveys (DHS) to examine households' waste management in LAC. DHS surveys are nationally representative household surveys that collect data on various aspects of health, including reproductive health, child health, and nutrition. They cover both low- and middle-income countries, enabling comparative analyses across different contexts. Moreover, these surveys provide detailed information on garbage collection, such as type and frequency, which is relevant for our hypothesis that children's health outcomes are negatively correlated by higher exposure to residential waste.

We retrieved garbage collection data from 14 surveys with a total of 118,597 mother-child dyads.¹¹ Access to garbage collection services varies significantly across countries. In Guyana, for instance, access to garbage collection was low at 12% in 1998, rising to 21% in 2005. Meanwhile, Nicaragua reported a low access rate of around 23% in both 1998 and 2001. The Dominican Republic showed progress, with the percentage of access increasing from 61% in 2007 to 75% in

¹¹ See Appendix Table A5 for additional details.

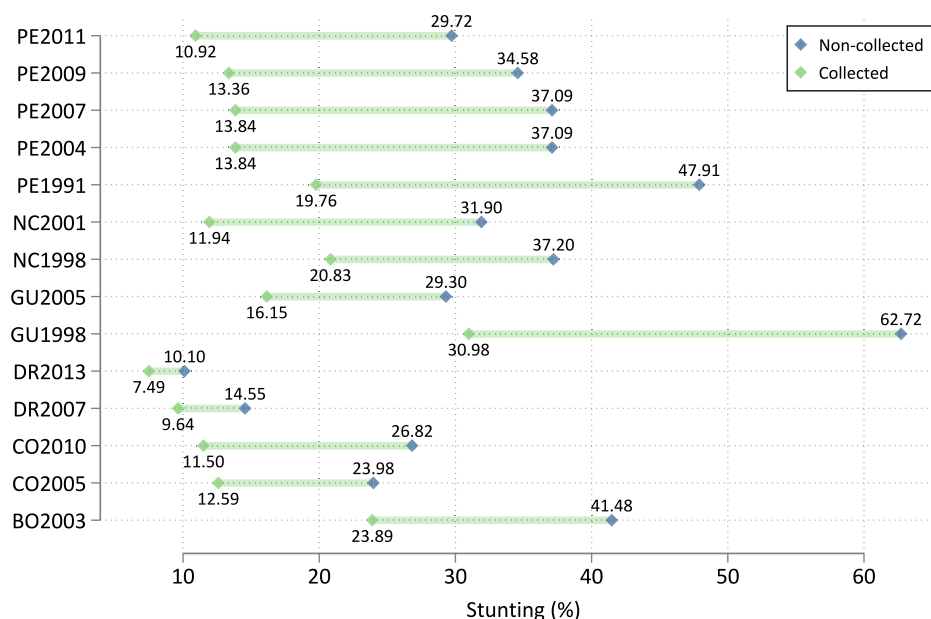


Fig. 4. Stunting by type of garbage disposal system.

Notes: This figure presents the percentage of children with stunting across different surveys by access to garbage collection. Data were obtained from the Demographic and Health Surveys (DHS) for Bolivia (BO), Colombia (CO), Dominican Republic (DR), Guyana (GU), Nicaragua (NC), and Peru (PE). The data presented here only include countries for which garbage collection information is available from the DHS surveys. Statistics represent unweighted averages.

2013. Notably, access to garbage collection services is steady in Peru across multiple surveys from 1991 to 2011. Even over two decades, the percentage of access to garbage collection services remained almost stagnant, hovering around 34%. Additional statistics for the pooled sample of countries/surveys are available in Appendix Table A6.

Fig. 4 presents differences in stunting by the availability of garbage collection across surveys. The largest difference in stunting between children with and without garbage collection services is observed in Guyana (1998), at 31% and 63%, respectively, while the smallest difference in stunting was observed in the Dominican Republic (2013), at 7% and 10% respectively.

4. Results

In this section, we provide evidence of the potentially significant role that access to basic infrastructure services, such as water, sanitation, and garbage collection, plays in early childhood health and development, in addition to the clear and compelling statistics we just presented in Section 2. Specifically, the MICS data demonstrate a strong link between access to water and sanitation and children's likelihood of being on track in their development, according to the ECDI, as well as a lower likelihood of stunting. In addition, the DHS data highlight a negative association between access to improved garbage collection services and rates of stunting among children under five.

Water and sanitation infrastructure using the MICS: We present the results of the linear probability models that explore the association between access to water and sanitation and developmental and health outcomes. In Table 1, we analyze the association between access to water and sanitation, and a child's developmental outcomes as measured by the ECDI. The data utilized for this analysis originate from 22 surveys conducted across 14 different countries. It is important to note that the specific countries and surveys included in the analysis vary based on the model and health outcome in question, as not all outcomes are available across all surveys.¹² The models, spanning

columns 1 to 8, reveal a consistent positive association. For instance, the model in column 1, accounting for — year fixed effects, suggests that access to water correlates with a 3.9% increase in the likelihood of a child being developmentally on track according to the ECDI. This relationship persists even when additional control variables are introduced, including child, mother, and household characteristics. The coefficients for water and sanitation access slightly diminish but remain significantly positive, emphasizing the robustness of these findings. Children with water access exhibit a 1.9% to 4.1% higher likelihood of optimal development, while sanitation access corresponds to a 2.9% to 5.8% increase.

In Appendix Table A8, we refine this analysis, dissecting the ECDI into its constituent sub-domains: learning, socio-emotional development, physical development, and literacy/numeracy. The models reveal that water access predominantly benefits learning, socio-emotional development, and literacy/numeracy, with associations as high as 7.6%, 1.5%, and 1.9% respectively. Sanitation access echoes these benefits, particularly enhancing learning (9.0%), socio-emotional development (2.0%), and literacy/numeracy (3.4%). The physical development sub-domain, while showing a positive trend, does not reach statistical significance, possibly due to short-term measurement limitations or variability in physical development milestones.

In Table 2, we explore the associations between access to water and sanitation and health outcomes, particularly stunting. The analysis reveals a noteworthy correlation: access to water is associated with a 1.3% to 3.8% reduction in the likelihood of stunting. These associations are further strengthened with access to sanitation, which is correlated with a 4.8% to 8.1% decrease in the likelihood of stunting. Although the percentage changes might seem modest, they are significant when considered in relation to the baseline prevalence of these health outcomes, thus highlighting their potential relevance for public health efforts.

We tested additional models that include wealth to mitigate endogeneity issues and obtain more accurate estimates. Specifically, wealth

¹² The sample across the different models and outcomes is not harmonized due to substantial variations in the number and type of observations when

comparing Tables 1 and 2. Despite this, the results remain consistent and robust when using the same sample, as evidenced by the findings presented in Appendix Table A7.

Table 1
Access to infrastructure and the relationship with the ECDI.

Dependent variable	ECDI (dummy)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Water access (dummy)	0.039*** (0.009)	0.041*** (0.008)	0.027*** (0.008)	0.019** (0.009)				
Sanitation access (dummy)					0.059*** (0.009)	0.058*** (0.009)	0.037*** (0.009)	0.029*** (0.009)
R2-squared	0.039	0.051	0.059	0.063	0.040	0.052	0.060	0.063
Mean of outcome	0.834	0.834	0.834	0.834	0.834	0.834	0.834	0.834
N	37 964	37 964	37 964	37 964	37 964	37 964	37 964	37 964
Clusters	10 385	10 385	10 385	10 385	10 385	10 385	10 385	10 385
<i>Controls</i>								
Country–region–year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Children level	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Mother level	No	No	Yes	Yes	No	No	Yes	Yes
Household characteristics	No	No	No	Yes	No	No	No	Yes

Notes: This table presents results from a linear probability model for the correlation between child development indicators and access to water and sanitation infrastructure. The analysis sample is composed from 19 surveys included in The Multiple Indicator Cluster Surveys (MICS). The countries that have been included in this analysis and their respective sample sizes can be found in Appendix Table A1. The analysis controls for fixed effects at the country–region–year level. Additional controls at the child level include age (in months), birth cohort, and sex. Mother-level controls account for the mother's education level. Household characteristics controlled for are urban residency status, access to electricity, total household size, and the number of household members who are under five years of age. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Table A4. Primary sampling unit (PSU) clustered standard errors are reported in parentheses (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2
Access to infrastructure and the relationship with health outcomes.

Dependent variable	Stunted (dummy)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Water access (dummy)	−0.038*** (0.006)	−0.038*** (0.006)	−0.026*** (0.006)	−0.013** (0.006)				
Sanitation access (dummy)					−0.081*** (0.006)	−0.081*** (0.006)	−0.061*** (0.006)	−0.048*** (0.006)
R2-squared	0.044	0.050	0.060	0.068	0.047	0.053	0.062	0.069
Mean of outcome	0.116	0.116	0.116	0.116	0.116	0.116	0.116	0.116
N	67 248	67 248	67 248	67 248	67 248	67 248	67 248	67 248
Clusters	8190	8190	8190	8190	8190	8190	8190	8190
<i>Controls</i>								
Country–region–year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Children level	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Mother level	No	No	Yes	Yes	No	No	Yes	Yes
Household characteristics	No	No	No	Yes	No	No	No	Yes

Notes: This table presents results from a linear probability model for the correlation between child development indicators and access to water and sanitation infrastructure. The analysis sample is composed from the surveys included in the Multiple Indicator Cluster Surveys (MICS). The countries that have been included in this analysis and their respective sample sizes can be found in Appendix Table A1. The analysis controls for fixed effects at the country–region–year level. Additional controls at the child level include age (in months), birth cohort, and sex. Mother-level controls account for the mother's education level. Household characteristics controlled for are urban residency status, access to electricity, total household size, and the number of household members who are under five years of age. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Table A6. Primary sampling unit (PSU) clustered standard errors are reported in parentheses (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

may predict both access to improved sanitation facilities as well as developmental and health outcomes, leading to issues of selection or omitted variable bias. By controlling for wealth, we can account for the possibility that wealthier individuals are more likely to have better sanitation facilities while also simultaneously enjoying better health outcomes due to other factors that are correlated with wealth. These results are shown in Appendix Table A9. In our baseline model, we have opted not to include wealth in our main estimations due to the construction of the wealth index, which integrates proxies related to infrastructure access. This integration could potentially mask the independent effects of water and sanitation access.¹³

The estimates show that even after controlling for household wealth, water and sanitation access are both positively associated with on-track ECDI scores and negatively associated with stunting. However, only sanitation is statistically significant. Previously, all our models control for maternal education in order to separate out the impact of maternal education on child developmental and health outcomes. Indeed, maternal education is linked to better hygiene, nutrition, and

healthcare-seeking behavior for children, and may influence household wealth and investment in water and sanitation infrastructure. Lastly, we also tested the robustness of the results to the inclusion of the stimulation index, which is a strong predictor of ECD (Cuartas et al., 2023), obtaining consistent estimates.

To further validate our findings, we conducted additional exercises that included a broader set of control variables and alternative model specifications as discussed in Section 2. We incorporated recent illness episodes, vaccination behavior, and health status as additional controls, recognizing their potential influence on child development and health outcomes (Appendix Table A9). By including these variables, we aimed to capture the immediate health environment of the child, which could confound the relationship between infrastructure access and developmental outcomes. These controls help ensure that the observed associations are not driven by recent health shocks or varying levels of preventive health behaviors across households. Additionally, we examined the impact of community-level characteristics in our estimates by using PSU fixed effects (Appendix Tables A9 and A10). By incorporating PSU fixed effects, we control for unobserved heterogeneity at a more granular level, capturing community-specific factors that could influence the outcomes. This approach allows us to address potential confounding from both observed and unobserved

¹³ See Appendix Table A2 for additional details about the construction of the wealth index.

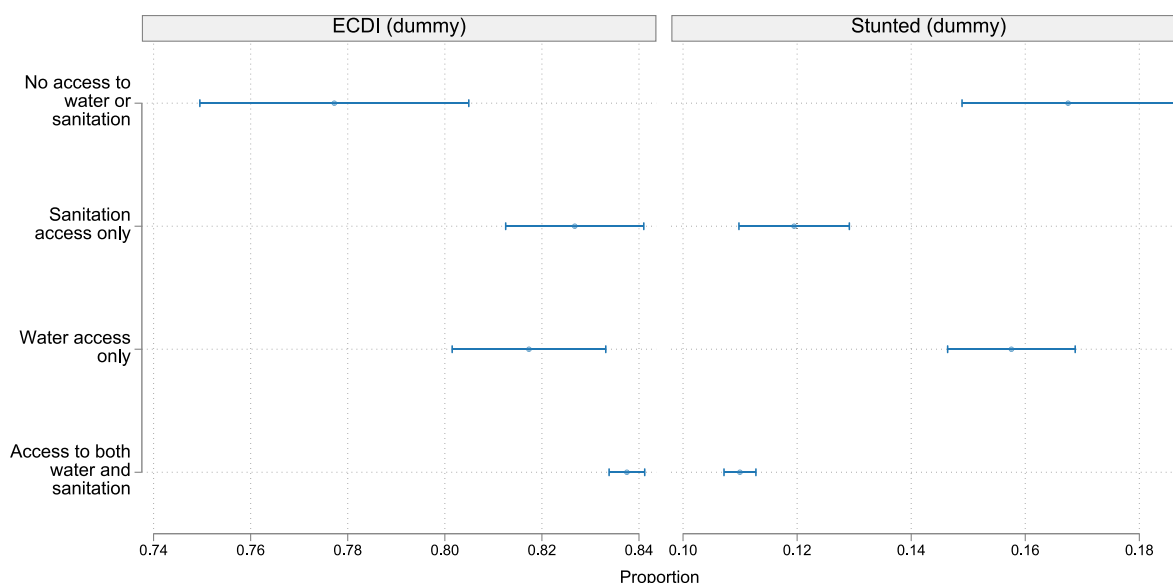


Fig. 5. Predicted developmental and health outcomes by infrastructure access.

Notes: The coefficient plot visually displays the predictive margins for three distinct health and development outcomes in children: being on track according to the early childhood development index (ECDI) and being stunted. The figure categorizes the results based on the children's access to water and sanitation, delineating four scenarios: "No access to water or sanitation", "Water access only", "Sanitation access only", and "Access to both water and sanitation". The predictive margins are calculated from the linear probability models in Appendix Table A11. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Tables A3 and A4. Confidence intervals are represented by capped lines and set at 90%.

community-level characteristics more accurately. The results are robust to these alternative specifications.

When recognizing the inherent interconnectedness of sanitation and water in determining health and developmental outcomes, it becomes essential to investigate not just their individual influences, but also their combined effects. The results of these interactions are illustrated in Fig. 5. By including an interaction term between sanitation and water, we aim to distinguish whether their joint influence amplifies or diminishes the outcomes observed when they act independently. The results for ECDI indicate a positive association with both water and sanitation access. Children with access to both water and sanitation have a predicted 83.8% likelihood of being on track in ECDI, which is higher compared to those with access to only water (81.7%), only sanitation (82.7%), or neither (77.7%). When it comes to stunting children with access to both water and sanitation have the lowest likelihood of being stunted, at 11.00%. Those without access to either have a higher likelihood of 16.75%. Similar trends are observed for children with access to only water (15.76%) or only sanitation (11.95%).

In order to mitigate the negative impacts of inadequate access to clean water and sanitation facilities, households may adopt various behaviors or attitudes that promote health and well-being. These practices, ranging from treating water to enhance its safety for consumption, to ensuring the availability of soap or cleansers for handwashing, play a crucial role in safeguarding health, particularly in contexts where access to water and sanitation infrastructure is limited. As shown in Fig. 6(a), water treatment emerges as a significant mitigating factor, especially for developmental outcomes. Promoting household water treatment practices could be a viable interim solution while working towards universal access to water and sanitation. The data also suggest that the availability of a cleanser might play a mitigating role, particularly in relation to ECDI and stunting (see Fig. 6(b)).

The robustness of our results is reinforced through a series of additional analyses. First, we confirm our results through the application of a preferential within propensity score matching approach (Arpino & Cannas, 2016), as demonstrated in Appendix Table A13, providing further confidence in the validity and reliability of our findings. Second, we extend further our analyses with the results presented in Tables A14 and A15. By employing a categorical variable for both water and

sanitation access, rather than relying on binary indicators, we explore the correlation of different types of access. Our findings interestingly highlight the significant role of packaged water in mitigating the potential adverse effects associated with other forms of water access. This observation could be indicative of the unreliable quality of piped water sources in some Latin American countries and the lack of appropriate storage options, underscoring the potential need for alternative safe drinking water solutions. On the sanitation front, flush systems stand out as particularly beneficial, showcasing greater effectiveness in promoting health and developmental outcomes when compared to both hygienic non-flush methods and unsanitary systems, as well as in comparison to a complete lack of sanitation facilities. These findings underline the necessity to recognize and address the varying impacts of different types of water and sanitation access. They also emphasize the importance of considering the quality and type of water and sanitation services alongside their availability.

The data from the MICS compellingly illustrate the positive relationship that investments in water and sanitation infrastructure can have on the development and health of young children. This relationship holds strong even when taking into account various individual, maternal, and household factors, alongside considerations of household wealth and stimulation activities, as well as recent illness episodes, vaccination behavior, health status, hygiene practices, and PSU fixed effects to control for unobserved community-level heterogeneity. To get more into this association, we expand the data analyses to include information on residential waste collection systems. Such data would provide a more comprehensive understanding of how waste management infrastructure services impact early childhood development. Since exposure to waste can affect the quality of an individual's environment, this additional information may enhance the accuracy of the relationship between infrastructure services and early childhood development.

Residential waste management infrastructure using the DHS: Table 3 presents the results of the linear probability model to assess the correlation between garbage collection and child health indicators, specifically stunting. Our analysis begins with a simple correlation using year-fixed effects, which is then extended by introducing additional controls, including children, mother and household characteristics. Controlling for household characteristics allows us to account for the potential

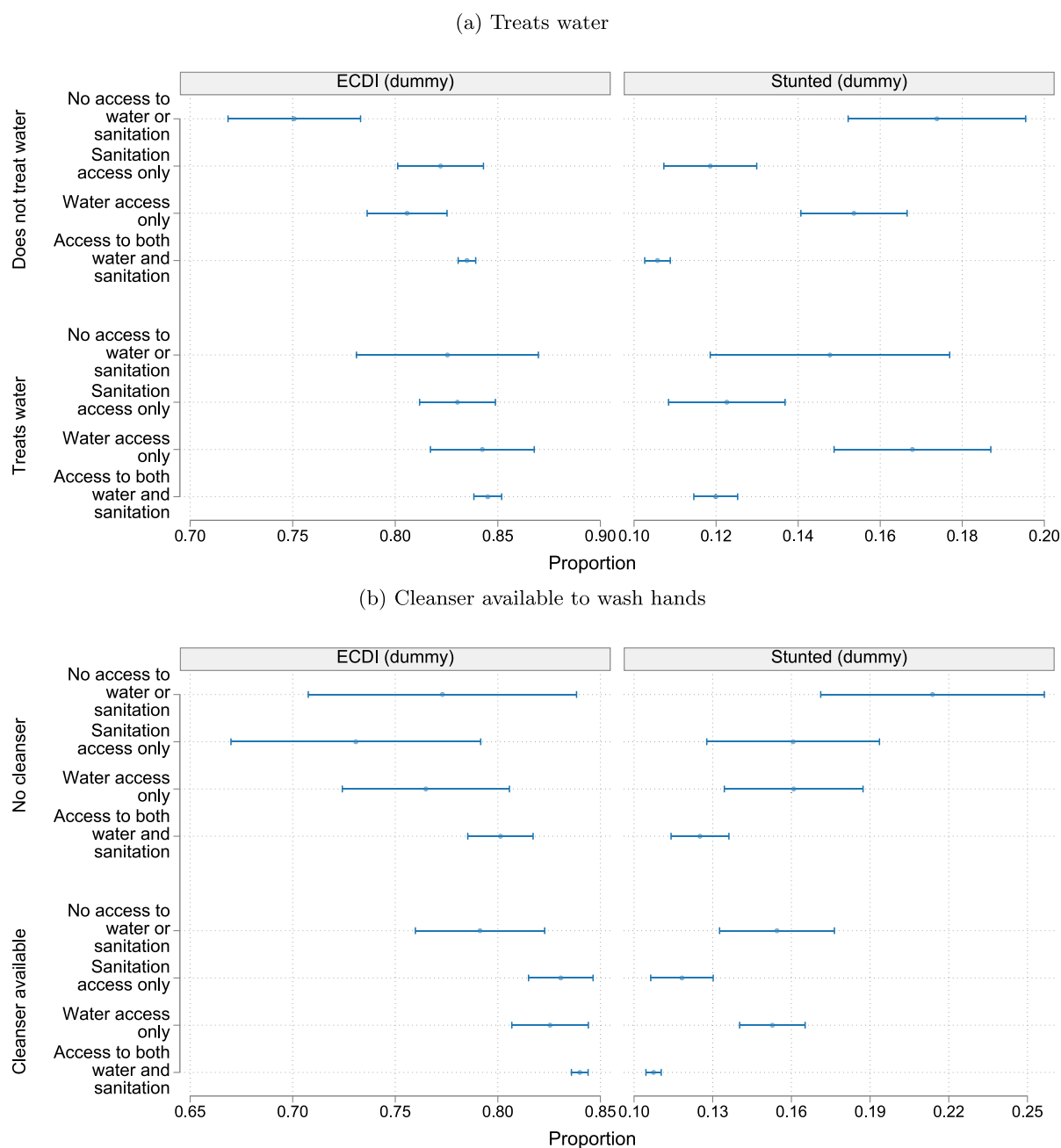


Fig. 6. Predicted developmental and health outcomes by mitigation behavior and infrastructure access.

Notes: The coefficient plot visually displays the predictive margins for three distinct health and development outcomes in children: being on track according to the early childhood development index (ECDI) and being stunted. The results are categorized based on the household's adoption of mitigation behaviors such as treating water before consumption to make it safe (Fig. 6(a)) and having hand cleanser available at home (Fig. 6(b)). The figure then presents estimations according to the children's access to water and sanitation, outlining four scenarios: "No access to water or sanitation", "Water access only", "Sanitation access only", and "Access to both water and sanitation". The predictive margins are calculated from the linear probability models in Appendix Table A12. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Tables A3 and A4. Confidence intervals are represented by capped lines and set at 90%.

impact of socio-economic status on the relationship between garbage collection and health indicators, as discussed previously.

The results indicate a statistically significant association between access to improved garbage collection services and child development indicators. Children under five with access to improved garbage collection services have a 13.4% lower probability of stunting in the most straightforward model. These coefficients are lower, however, when additional control variables are included. Specifically, results from the model including all covariates show a statistically significant association between improved garbage collection and the health outcomes. Overall, the coefficients for stunting range from -0.134 to

-0.028 , representing approximately as much as 50% of the mean of the dependent variable. Our results remain consistent when controlling for wealth and interacting it with access to garbage collection, however, children from poorer households are more likely to suffer from stunting — see Appendix Figure A2. We conducted additional exercises that included a broader set of control variables and alternative model specifications, similar to the approaches used in the MICS analysis. We incorporated recent illness episodes, vaccination behavior, and health status as additional controls (Appendix Table A16). Additionally, we examined the impact of community-level characteristics by using PSU fixed effects (Appendix Tables A16 and A17). The results are robust

Table 3
Access to waste management infrastructure and the relationship with health outcomes.

Dependent variable	Stunted (dummy)			
	(1)	(2)	(3)	(4)
Garbage collection (dummy)	−0.134*** (0.003)	−0.134*** (0.003)	−0.091*** (0.003)	−0.028*** (0.004)
R2-squared	0.123	0.145	0.165	0.181
N	118 712	118 712	118 712	118 712
Clusters	17 568	17 568	17 568	17 568
Mean of outcome	0.251	0.251	0.251	0.251
<i>Controls</i>				
Country–region–year FE	Yes	Yes	Yes	Yes
Children level	No	Yes	Yes	Yes
Mother level	No	No	Yes	Yes
Household characteristics	No	No	No	Yes

Notes: This table presents results from a linear probability model for the correlation between child health indicators and access to water and sanitation infrastructure. The analysis sample is composed from fourteen surveys included in the Demographic and Health Surveys (DHS). The countries that have been included in this analysis and their respective sample sizes can be found in Appendix Table A1. The analysis controls for fixed effects at the country–region–year level. Additional controls at the child level include age (in months), birth cohort, and sex. Mother-level controls take into account the mother's education level and whether she is the head of the household. Household characteristics include urban residency status, access to electricity, water, sewage, total household size, and number of members under five years old. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Table A6. Primary sampling unit (PSU) clustered standard errors are reported in parentheses (***) $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

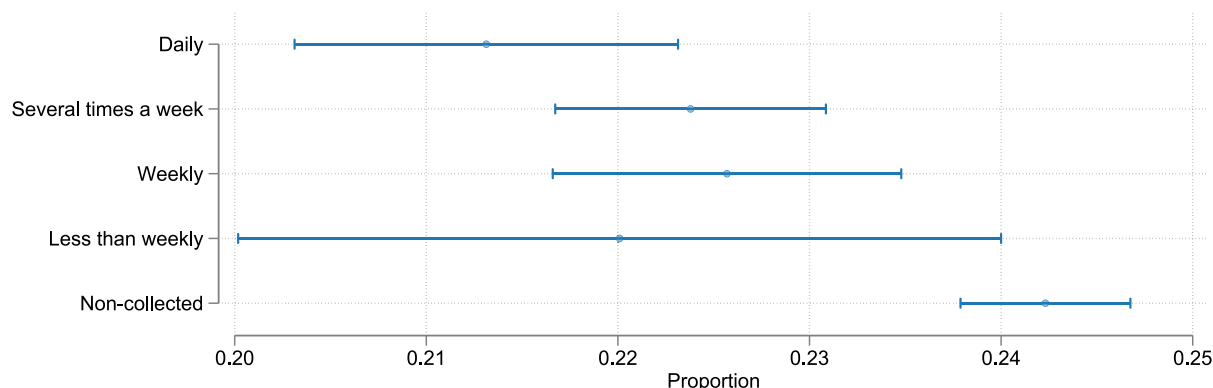


Fig. 7. Predicted stunting with changes in frequency of garbage collection.

Notes: The coefficient plot visually displays the predictive margins for children being stunted. The figure presents estimations according to the children's access to garbage collection and frequency. The predictive margins are calculated from the linear probability models in Appendix Table A19. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Table A6. Confidence intervals are represented by capped lines and set at 90%.

to these alternative specifications, reinforcing the association between improved garbage collection services and better health outcomes for children.

We also analyze a sub-sample of surveys that include the frequency with which garbage collection services are provided. In particular, we distinguish between households that benefit from garbage collection daily, several times a week, weekly, and less than weekly. The results in Fig. 7 show that children living in households with daily garbage collection have lower rates of stunting compared to those living in households where garbage is not collected. The main differences appear to stem from the gap between these two groups.

In Appendix Figure A3, we summarize the findings of an analysis that examines the potential interaction between garbage collection and access to electricity, water, and sanitation services in relation to child development indicators.¹⁴ Specifically, the figure shows predicted average rates of stunting for children under five, listed separately for households with and without access to these services and improved garbage collection. The data indicate that access to improved garbage collection is consistently associated with lower stunting rates across different infrastructure scenarios, regardless of electricity, water, or

sanitation services availability. Households with access to all three services (electricity, water, and sanitation) and garbage collection have substantially lower stunting rates than households without adequate garbage collection. This finding suggests that waste management infrastructure plays a distinct and essential role in reducing stunting, even when other services are present.

We further analyze the predicted rates of stunting in households with and without access to garbage collection, looking at certain household behaviors that could reduce children's environmental exposure to waste-related impacts. In particular, we consider households with and without a refrigerator, which could potentially minimize food spoilage and the associated health risks. We then examine the use of bottled drinking water, which may reduce the likelihood of water contamination from improper waste disposal. Finally, we investigate whether households store their garbage in lidded containers before disposal, a practice that could mitigate the spread of disease-carrying vectors and decrease children's exposure to harmful waste.

As shown in Fig. 8, no differences are detectable in the likelihood of stunting between children in households with refrigerators, regardless of whether the households have garbage collection. But, among children in households without a refrigerator, those who are without garbage collection have higher rates of stunting (4.60 percentage points). In Figure A4, we show the variation in rates of stunting

¹⁴ See Appendix Table A5 for additional details.

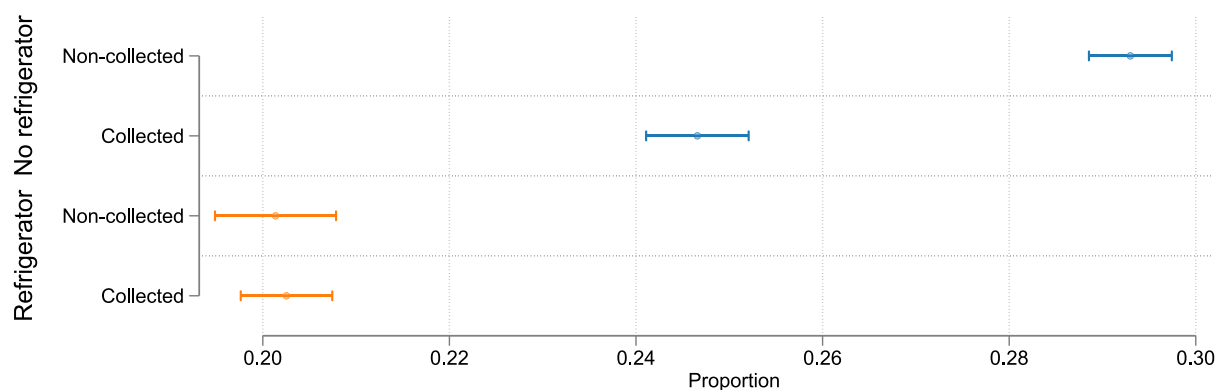


Fig. 8. Predicted stunting in households with refrigeration ownership and type of garbage collection.

Notes: The coefficient plot visually displays the predictive margins for children being stunted. The results are categorized based on the household's ownership of refrigerators. The figure then presents estimations according to the children's access to garbage collection. The predictive margins are calculated from the linear probability models in Appendix Table A21. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Table A6. Confidence intervals are represented by capped lines and set at 90%.

based on households' main source of drinking water and availability of garbage collection. The results indicate that households without garbage collection have higher rates of stunting than households with garbage collection when these households use piped or natural sources as their main source of drinking water, although the latter estimate is more imprecise due to the small number of households that rely on natural sources.

Fig. 9 reveals that households that store waste in lidless containers had the biggest gap in stunting by garbage collection access. Specifically, we observe that households using lidless containers had a predicted stunting rate of 27.99% when garbage was not collected, compared to 23.71% when garbage was collected. Similar differences remain for households that store their waste in vessels other than covered containers, such as plastic bags. These findings suggest that the use of lidded containers for waste can have significant benefits for promoting healthy child development. By reducing exposure to potential pollutants and promoting sanitary waste management practices, households can help protect their children. Furthermore, these results highlight the importance of targeted policy interventions aimed at promoting the use of lidded containers for waste among households in the region.

We further extend our analyses to look more closely at the relationship between child health and waste management by examining the predicted rates of stunting among children who have experienced recent illness. Our findings are summarized in Figure A5. Specifically, we consider instances of child diarrhea, cough, and fever, as reported by households during the survey period. This analysis sheds light on the potential mechanisms underlying the observed associations, particularly emphasizing a health channel. We find that children living in households with garbage collection have a lower likelihood of being stunted than children in households without garbage collection, independently of recent illness episodes.

5. Conclusion

This paper provides insights into the link between access to essential infrastructure services and early childhood development (ECD) in Latin America and the Caribbean (LAC). Our findings show that access to water, sanitation, and garbage collection can significantly impact children's developmental and health outcomes. Specifically, we document that access to water and sanitation is strongly associated with a higher likelihood of being developmentally on track as measured by the early childhood development index and with fewer instances of stunting. Additionally, the results reveal a negative association between improved garbage collection services and rates of stunting among children under five. Our findings hold strong even when taking into account various

individual, maternal, and household factors, alongside considerations of household wealth and stimulation activities.

Our findings are consistent with prior evidence (Gao et al., 2021; Piper et al., 2017; Schady, 2015; WaterAid, 2016), and have important implications for policymakers in the region. First and foremost, the expansion of piped water and sanitation services is crucial for improving child outcomes. This expansion aligns with evidence indicating that improvements in water infrastructure can substantially reduce child mortality, with stronger associations among socioeconomically disadvantaged groups. It is essential, however, to recognize the complexity of separating the effects of improved water from that of improved sanitation, as the magnitude of the relationship of clean water with child outcomes is often different than that of improved sanitation.

A key dimension of our analysis that merits significant emphasis is the differential exposure of types of access to infrastructure, a crucial aspect considering the Sustainable Development Goals (SDGs) have elevated standards for what constitutes adequate access to water and sanitation. Our analysis underscores the pivotal role of alternatives to piped water in either mitigating or exacerbating the potential effects of water quality received by households. This is especially pertinent in contexts where piped water sources may be unreliable, highlighting the need for alternative safe drinking water solutions. Regarding sanitation, flush systems demonstrate substantial benefits, promoting health and developmental outcomes more effectively than both hygienic non-flush methods and unsanitary systems, even in the absence of sanitation facilities. These findings stress the need to recognize and address the varying impacts of different types of water and sanitation access, emphasizing the necessity for specific measurements of quality and access to essential infrastructure. There exist challenges in accurately measuring real household access, a critical consideration for policy and infrastructure development.

We underscore the importance of investing in waste management infrastructure as a critical but often overlooked component of comprehensive strategies to enhance early childhood development in low- and middle-income countries. This is particularly critical in light of previous work (Lu et al., 2020; McCoy et al., 2022) and our descriptive observations of persistent inequities in access to basic water, sanitation, and waste disposal services in LAC. By ensuring that all households have access to all basic services that promote good health and foundational skill development, policymakers can improve the developmental trajectories and well-being of children across the region and maximize the beneficial impact of other investments (including early childhood care and education and parenting programs) while promoting sustainable economic development for years to come.

Our paper also highlights the need for targeted policy interventions aimed at improving household practices in the region. The literature

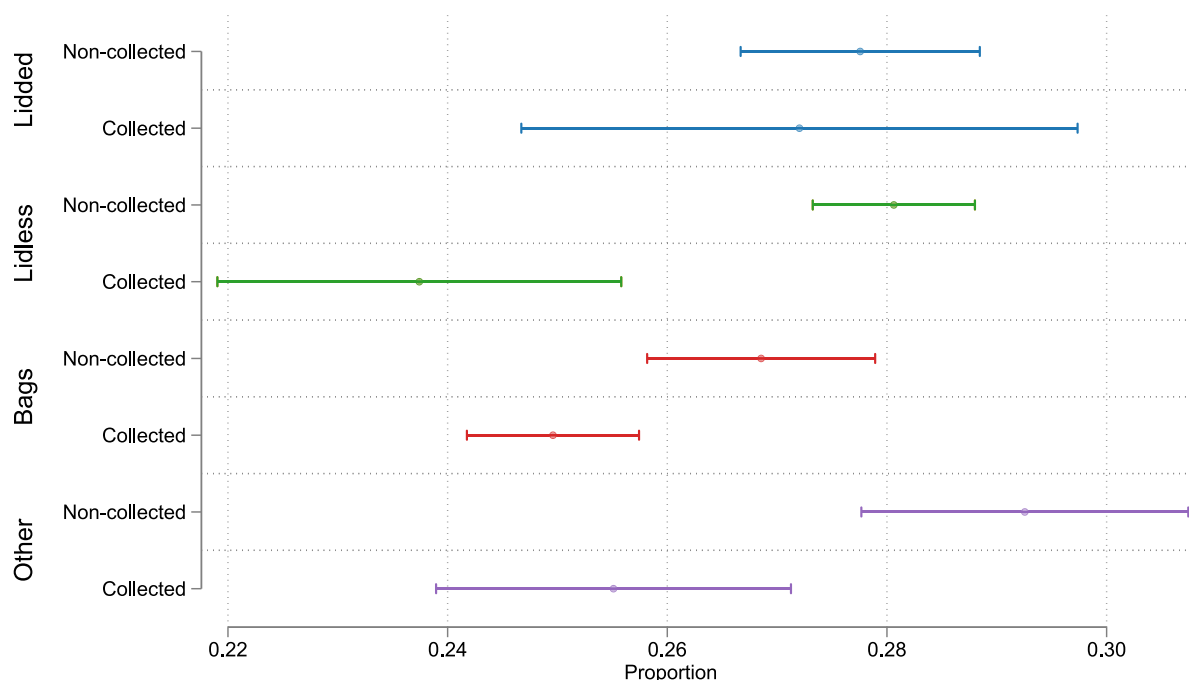


Fig. 9. Predicted stunting in households using lidded containers for waste by type of garbage collection.

Notes: The coefficient plot visually displays the predictive margins for children being stunted. The results are organized according to the type of garbage containers used by households prior to collection. The figure then presents estimations according to the children's access to garbage collection. The predictive margins are calculated from the linear probability models in Appendix Table A23. Variable definitions are available in Appendix Table A2 and summary statistics in Appendix Table A6. Confidence intervals are represented by capped lines and set at 90%.

focuses on infrastructure hardware rather than on interventions that modify household behaviors and mitigation measures (Rupasinghe, Benton, Mandeville, & Latypova, 2021; Schady, 2015). For example, households could be encouraged to protect their environment from potential pollutants through mitigation measures such as storing waste in lidded containers or participating in community-based recycling programs. Similarly, strategies like handwashing, and treating water hold promise, but their effectiveness often relies on behavioral changes. Policymakers could consider implementing public education campaigns to raise awareness of the importance of proper household practices for healthy child development. Parenting and ECD programs delivered through home visits and community meetings have been expanding in LAC and offer a potential means of providing information and support related to hygiene practices and the protection of children from environmental hazards (Leer & Lopez-Boo, 2019).

Importantly, our findings suggest that households in the region can reap significant economic benefits from investments in water, sanitation, and waste management infrastructure and their downstream benefits for child health and development. Indeed, recent simulation studies estimate that stunting alone causes learning and economic losses with a total cost of approximately \$176.8 billion per birth cohort globally and \$44.7 billion in Latin America (Fink et al., 2016), while the cost of not investing in critical interventions to promote ECD in LAC countries such as Guatemala, Nicaragua, Colombia, Peru, Ecuador, and Chile ranged from 0.05% to 3.6% of their gross domestic product (Richter et al., 2017). The cost of inaction is therefore substantial and has meaningful impacts on individual and societal outcomes. By promoting ECD and reducing rates of stunting among young children, practices to improve water, sanitation, and waste management can lead to lower healthcare costs for families, maximize the impact of subsequent interventions, and promote higher levels of educational attainment and workforce productivity over time. There is also a necessity for a multi-sector approach. We underline the need to jointly address a range of factors — environmental, nutritional, health-related — to enhance early childhood development effectively (Black et al., 2017; Rupasinghe et al., 2021; Vargas-Baron et al., 2019).

While this study makes significant contributions to the literature, it has limitations that inform future research. First, the estimated associations between water, sanitation, garbage collection, and ECD were robust to different model specifications and the inclusion of key control variables; Yet, it is impossible for us to control for all potential confounders. Therefore, future research could employ longitudinal data and experimental or quasi-experimental designs to provide further evidence on the potential causal effects of access to critical infrastructure on ECD. Future research could also explore the specific mechanisms through which access to essential infrastructure services affects child health outcomes, as well as the potential role of cultural and social factors in shaping household practices and moderating the observed associations in the region. Additionally, further studies could explore the long-term economic benefits of investing in infrastructure for promoting healthy child development and sustainable economic growth.

CRedit authorship contribution statement

Lenin H. Balza: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jorge Cuartas:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nicolas Gomez-Parra:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tomás Serebrisky:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.worlddev.2024.106817>.

Data availability

Data will be made available on request.

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